Automated Storage & Retrieval System

Thesis submitted in partial fulfilment

of the requirements of the degree of

Masters in Science

by

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Under the Supervision of

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April 2023

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Malad, Mumbai 400064

CERTIFICATE

This is to certify that the dissertation entitled **"Automated Storage & Retrieval System"** is a bonafide work of "**Punshi Barot" (Roll no: 4 and G.R no: 3480578)** submitted to the Nagindas Khandwala College(Autonomous),Mumbai in partial fulfillment of the requirement for the award of the degree of **"Masters in Science"**.

Prof. Rashmi Tiwari

Internal-Examiner

External Examiner

Supervisor's Certificate

This is to certify that the dissertation entitled **"Automated Storage & Retrieval System"** submitted by **PUNSHI RANSHI BAROT, Roll no: 4, G.R no: 3480578,** is a record of original work carried out by him/her under my supervision and guidance in partial fulfillment of the requirements of the degree of **Masters in Science** at Nagindas Khandwala College(Autonomous),Mumbai 400064 . Neither this dissertation nor any part of it has been submitted earlier for any degree or diploma to any institute or university in India or abroad.

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APPOINTMENT LETTER

Dear Punshi 15th Oct 2021

Sub: Appointment Letter

Congratulations and Welcome aboard!!!!

We are pleased to inform that you have been appointed for the post of Software Engineer in Halston Automation and would be required to join on **18th Oct 2021**

Halston is a specialist provider of Solutions for Managing Large-Scale, Distributed and Hybrid Networks.

We are leading supplier and integrator of systems and solutions for the industrial and Process automation industry. Our team of veteran engineers have commissioned critical projects for major accounts across the globe by locally developing and integrating technologically advanced software solutions.

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Name: Punshi Barot
Designation: Software Engineer
Location: Mumbai

Provisional Salary: 55,000 / Month (3 Months)
Revised Salary after completion of 3 Months: Rs 70,000 / Month

Salary would be done on 1st of Every Month

Allowance: Below are various allowance that would be provided

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Halston has over the years maintained the informal office environment and Team effort for the projects. I am sure, as everyone else, you too will enjoy the experience that you would get at Halston.

Mihir Shah

(CEO)

Declaration of Originality

I, ***Punshi Barot*, Roll no: 4, G.R no: 3480578**, hereby declare that this dissertation entitled "***Automated Storage & Retrieval System"*** presents my original work carried out as a Master Student of Nagindas Khandwala College(Autonomous),Mumbai 400064. To the best of my knowledge, this dissertation contains no material previously published or written by another person, nor any material presented by me for the award of any degree or diploma of Nagindas Khandwala College(Autonomous),Mumbai or any other institution. Any contribution made to this research by others, with whom I have worked at Halston Automation, Mumbai or elsewhere, is explicitly acknowledged in the dissertation. Works of other authors cited in this dissertation have been duly acknowledged under the sections "Reference" or "Bibliography". I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

I am fully aware that in case of any non-compliance detected in future, the Academic Council of Nagindas Khandwala College(Autonomous),Mumbai may withdraw the degree awarded to me on the basis of the present dissertation.

Date:

Place:

Punshi Barot

Acknowledgement

I remain immensely obliged to **Prof. Rashmi Tiwari**, for providing me with the idea of this topic, and for his invaluable support in garnering resources for me either by way of information or computers also his guidance and supervision which made this Internship/Project happen.

I would like to say that it has indeed been a fulfilling experience and I thank my company **Halston Automation** to give me the chance of working on this project.

Abstract

Automated storage and retrieval system (AS/RS) is one of the major material handling systems, which is widely used in distribution centers and automated production environments. AS/RSs have been utilized not only as alternatives to traditional warehouses but also as a part of advanced manufacturing systems. AS/RSs can play an essential role in modern factories for work-in-process storage and offer the advantages of improved inventory control and cost-effective utilization of time, space and equipment. Many issues and approaches related to the efficiency improvement of AS/RSs have been addressed in the literature. This chapter presents an overview of this literature from the past 40 years. It presents a comprehensive description of the current state-of-the-art in AS/RSs and discusses future prospects. The focus is principally on travel time estimates and different control policies such as dwell-point of the stacker crane, storage assignment, request sequencing and so on. In particular, this chapter will provide researchers and decision makers with an understanding of how to apply existing approaches effectively.

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List of Abbreviation

LU	Loading Unit
AS/RS	Automated Storage and Retrieval Systems
Bays	Place where the operators interact with the plant
Systore	Name of the WMS+WCS provided by System Logistics
Corridor	Space between compartments, racks where machine can move freely.

CHAPTER 1

INTRODUCTION

1.1 Problem Statement:

Storing and retrieving goods in a warehouse or distribution center is a common and traditional concept. Storage/warehouse systems are diverse, but there are two types: manual and automated in terms of operation. According to the Peerless Research Group, only 4% of them are highly automated (Logistics Management, 2019). The rest are mostly manual and half-automated, where operators move around the facility, lift each product by forklifts and pick up by hand. The main drawback of manual storage systems is that they are dependently labour-intensive and non-computerized. They involve many moving pieces and heavy machinery other than forklifts controlled by storage/warehouse workers.

1.2 Objectives of the project:

The objectives of the project Automated Storage and Retrieval Systems (ASRS) may vary depending on the specific requirements and goals of the organization or business that is implementing the system. However, some common objectives of ASRS projects may include:

Improved Efficiency: ASRS can help improve the efficiency of storage and retrieval processes, reducing the time and effort required to store and retrieve goods. This can lead to increased productivity and reduced labor costs.

Increased Accuracy: ASRS can improve the accuracy of inventory management, reducing errors and the need for manual checks. This can help prevent stockouts and overstocking, improving customer service and reducing waste.

Space Optimization: ASRS can help optimize storage space, reducing the need for additional storage facilities and the associated costs.

Enhanced Safety: ASRS can enhance safety in the workplace by reducing the need for manual handling of heavy or hazardous goods.

Better Inventory Management: ASRS can provide real-time inventory management, allowing for better control over stock levels and improved forecasting of future needs.

Improved Customer Service: ASRS can help improve customer service by reducing order processing times and improving order accuracy.

Cost Savings: ASRS can help reduce operational costs, such as labor and storage costs, resulting in significant cost savings over time.

Overall, the primary objective of ASRS projects is to improve the efficiency, accuracy, safety, and cost-effectiveness of storage and retrieval processes, while enhancing customer service and optimizing storage space.

Automated Storage and Retrieval System (AS/RS) work with the existing ERP solution, there are some AS/RS solution available which use Shared Database with ERP, this has some disadvantage like data corruption, it difficult to define and enforce clear boundaries between systems, accountability.

In this AS/RS system the data exchange will be done through files, we will setup a FTP server which will have the files, ASRS and ERP can send/receive files from the FTP server.

1.2 Scope of the project:

The project covers how automation works in general and how Automated Storage and Retrieval System (AS/RS) works in practise, thus understanding what the disadvantage of the warehouses without automation are, also what important AS/RS in the warehouse is.

CHAPTER 2

Literature Review

2. General Description

In Chapter 2 we see the theoretical background of automation, Automated Storage and Retrieval Systems (AS/RSs) in practice

2.1 Automation System

We see the definition of traditional warehouses and automation, and why we need automation in warehouses now.

2.1.1 Automation in General:

"Automation can be defined as the technology by which a process or procedure is accomplished without human assistance. It is implemented using a program of instructions combined with a control system that executes the instructions. To automate a process, power is required, both to drive the process itself and to operate the program and control system. Although automation is applied in a wide variety of areas, it is most closely associated with the manufacturing industries." (Groover, 2016, p. 75).

The term "automation" was originally invented in 1946 by an engineering manager at Ford Motor company. An automated system three distinct parts: Power, Program of instructions, Control System.

2.1.2 Automation in warehouses:

Storing and retrieving goods in a warehouse or distribution center is a common and traditional concept. Storage/warehouse systems are diverse, but there are two types: manual and automated in terms of operation. According to the Peerless Research Group, only 4% of them are highly automated (Logistics Management, 2019). The rest are mostly manual and half-automated, where operators move around the facility, lift each product by forklifts and pick up by hand. The main drawback of

manual storage systems is that they are dependently labour-intensive and non-computerized. They involve many moving pieces and heavy machinery other than forklifts controlled by storage/warehouse workers.

Inaccurate counts and misplacement might result in time-wasting and running out of essential items for new orders. Moreover, there is a possibility that employees encounter injuries as more than 5% of traditional warehouse workers were injured in 2016 (Conveyco, 2019).

Manual storage/warehouse systems, despite their simplicity and low-cost investments, cannot optimize the space and capacity of the warehouse regarding height due to limited reach of forklifts, thus occupying the considerably large land area. In manual storage systems, numerous products are usually stacked up one on top of another, which might reduce the quality of some at the bottom. Because of the lack of automation and computerization, it is very challenging to track the inflow and outflow of each product and its quantity. Therefore, each product transaction needs to be monitored continuously to maintain its availability and stocking levels. The tracking and monitoring information are not shared within any automated systems or programs, thus accumulating the wastes of time and labour and making the whole process more cumbersome.

Most businesses nowadays are still utilizing manual storage systems, despite their unproductivity. The initial investment and further maintenance costs are low; the training program is simple for employees. The manual system is especially appropriate for small businesses because of their simple product supply chain. For larger businesses, operating storages manually possibly gain a lot of short-term benefits in the early stage. However, the accumulation of errors and time-wasting will decelerate business development gradually. Moreover, storing heavy products in a traditional warehouse requires many forklifts and crane systems. All lifting and moving are handled by many human workers. Therefore, the possibility of causing a serious incident is high.[1]

2.2 Background of Automated Storage and Retrieval System (AS/RS):

In this Chapter we try to focus more on operations of AS/RS in real life

2.2.1 General Description:

Automated Storage and Retrieval Systems (ASRS) have been in use since the 1950s, although their development and use have become increasingly widespread in recent decades. ASRS is a computer-controlled system that uses machines to store and retrieve items from a warehouse or distribution center. The earliest forms of ASRS were based on cranes and conveyor belts, which were used to move items

within a warehouse. These early systems were limited in their capabilities and were often slow and prone to breakdowns. However, as technology evolved, new types of ASRS were developed that were faster, more reliable, and more flexible.

In the 1960s, automated storage systems began to be used in the automotive industry, where they were used to store and retrieve parts and components for assembly lines. By the 1980s, ASRS had become more common in a variety of industries, including food and beverage, pharmaceuticals, and retail.

In the 1990s and 2000s, advances in robotics, sensors, and computer software helped to further improve the capabilities of ASRS. Today, ASRS is used in a wide range of industries and applications, including manufacturing, warehousing, e-commerce, and distribution.

The benefits of ASRS include increased efficiency, improved accuracy, enhanced safety, better inventory management, and cost savings. ASRS is also becoming more advanced with the integration of artificial intelligence, machine learning, and the Internet of Things (IoT), which are helping to improve its capabilities even further.

2.2.2 Challenges of designing Automated Storage and Retrieval System (AS/RS):

Designing an Automatic Storage and Retrieval System (AS/RS) can be a complex task that involves many challenges. Here are some common challenges associated with designing AS/RS:

1. **System Integration**: AS/RS systems involve various subsystems such as conveyor systems, material handling equipment, inventory control software, and warehouse management software. Integrating these subsystems can be a significant challenge, as they may be designed by different manufacturers or suppliers.
2. **System Optimization**: The design of an AS/RS must optimize the use of space, time, and resources while minimizing costs. The design must take into account the types of products to be stored, their dimensions, weight, and other characteristics.
3. **System Capacity**: The design of an AS/RS must consider the capacity requirements of the system in terms of storage and retrieval. The capacity requirements can vary significantly depending on the type of product, the demand profile, and the operating environment.

4**. System Maintenance**: AS/RS systems are complex systems that require regular maintenance to ensure their proper functioning. Maintenance can be a significant challenge, as it requires specialized skills and knowledge.

5. **Safety and Security**: AS/RS systems must be designed with safety and security in mind. The system must be able to handle the storage and retrieval of products safely, and the system must be secure to prevent theft or unauthorized access.

6. **System Cost**: AS/RS systems can be costly to design and implement, with costs ranging from hundreds of thousands to millions of dollars. Designers must balance the cost of the system with the benefits it provides in terms of efficiency, productivity, and profitability.

CHAPTER 3

Methodology

The methodology for designing an Automated Storage and Retrieval System (AS/RS) typically involves the following steps:

Define Requirements: The first step is to define the requirements of the AS/RS system. This involves identifying the types of products to be stored, the storage capacity required, the desired level of automation, and the budget available.

System Design: The next step is to design the AS/RS system. This involves determining the layout of the system, selecting the appropriate equipment such as conveyors and cranes, and selecting the software that will control the system.

System Simulation: Once the design is complete, the AS/RS system can be simulated to test its performance. This involves creating a virtual model of the system and using software to simulate its operation. Simulation can help identify potential issues with the design and optimize the system's performance.

Installation: Once the design and simulation are complete, the AS/RS system can be installed. This involves assembling the equipment, installing the software, and testing the system to ensure that it is operating as expected.

Training: The next step is to train the employees who will be working with the AS/RS system. This involves teaching them how to operate the equipment, use the software, and troubleshoot any issues that may arise.

Maintenance: Finally, the AS/RS system must be maintained to ensure that it continues to operate at peak efficiency. This involves regular inspections, cleaning, and repairs as needed.

Overall, the methodology for designing an AS/RS system involves a combination of engineering design, software development, and operational planning. It is essential to follow a structured methodology to ensure that the AS/RS system meets the requirements of the business and operates reliably over the long term.

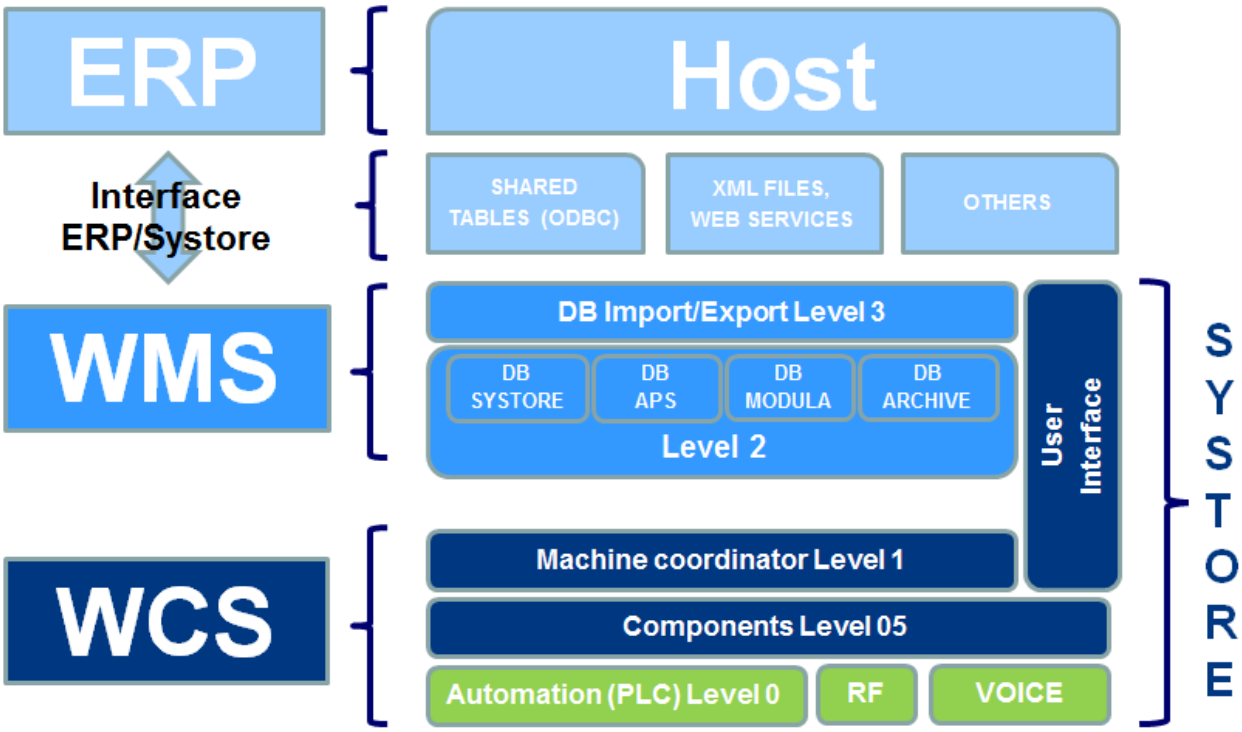


Fig 1: Multilevel Architecture

3.1 Core Code Segments:

The code defines the pallet dimensions like height, width and breadth.

```
/* Classi materiale */
insert into CLA_MATE (MATE_CLAMATE, MATE_DES, MATE_VAL, MATE_ERR)
values /*     VAL, ERR */
(1, 'Standard', 0,   0)
print formatmessage('CLA_MATE, Aggiunti %i records', @@rowcount)

/* Classi altezza
insert into CLA_ALT (ALT_CLAALT, ALT_DES, ALT_VAL, ALT_ERR)
values /*                 VAL, ERR */
(1400, 'Height 1400 mm', 1400,    0)
print formatmessage('CLA_ALT, Aggiunti %i records', @@rowcount)
*/

/* Classi larghezza */
insert into CLA_LARG (LARG_CLALARG, LARG_DES, LARG_VAL, LARG_ERR, LARG_MASK_0, LARG_MASK_1, LARG_MASK_2, LARG_MASK_3)
values /*                 VAL, ERR, MASK_0, MASK_1, MASK_2, MASK_3 */
(1000, 'Width 1000 mm',   1000,   0,     15,     15,     15,     15),
(1050, 'Width 1050 mm',   1050,   0,     15,     15,     15,     15)
print formatmessage('CLA_LARG, Aggiunti %i records', @@rowcount)

/* Classe lunghezza */
insert into CLA_LUNG (LUNG_CLALUNG, LUNG_DES, LUNG_VAL, LUNG_ERR, LUNG_MASK_0, LUNG_MASK_1, LUNG_MASK_2, LUNG_MASK_3)
values /*                 VAL, ERR, MASK_0, MASK_1, MASK_2, MASK_3 */
(1200, 'Length 1200 mm', 1200,   0,     15,     15,     15,     15),
(1250, 'Length 1250 mm', 1250,   0,     15,     15,     15,     15)
print formatmessage('CLA_LUNG, Aggiunti %i records', @@rowcount)
```

Fig 2: Defining pallet dimensions

```
/* Tipi Udc */
insert into TIPI_UDC (UDT_TIPOUDC, UDT_DES, UDT_PESOMAX, UDT_CLALUNG, UDT_CLALARG, UDT_CLAMATE,
 select UDT_TIPOUDC  = 'A'
       ,UDT_DES      = 'System Pallet 1050X1250X1400'
       ,UDT_PESOMAX  = 1205                    /* MAX PESO (KG) */
       ,UDT_CLALUNG  = 1250                    /* CLALUNG-DEBORDO  */
       ,UDT_CLALARG  = 1050                    /* CLALARG-DEBORDO */
       ,UDT_CLAMATE  = 1                       /* CLAMATE */
       ,UDT_MOVIM    = 1                       /* TIPOMOV */
       ,UDT_DIMX     = 1000                    /* DIMX */
       ,UDT_DIMY     = 150                     /* DIMY */
       ,UDT_DIMZ     = 1200                    /* DIMZ */
       ,UDT_OFFX     = 0                       /* OFFX */
       ,UDT_OFFY     = 0                       /* OFFY */
       ,UDT_OFFZ     = 0                       /* OFFZ */
       ,UDT_ALZO     = 0                       /* ALZO */
       ,UDT_NO_USER  = 0                       /* NO_USER */
       ,UDT_ID_IMG   = 1                       /* ID_IMG */
       ,UDT_ALT_MAX  = 1400                    /* CLASSE DI ALTEZZA */
       ,UDT_TM_LOAD  = 0                       /* TM_LOAD */
       ,UDT_USACENTR = 0                       /* USACENTR */
       ,UDT_CC_LATO  = 0                       /* CC_LATO */
       ,UDT_OK_EVAC  = 1                       /* OK_EVAC */

print formatmessage('TIPI_UDC, Aggiunti %i records', @@rowcount)
```

Fig 3: Assigning pallet dimensions

3.2 Requirements:

The AS/RS system requirement can be divided into 2 parts

1. Machinery which includes racks, shuttle, elevator.
2. Software which includes plc programming, database and GUI development

We will focus on database part of the AS/RS system.

3.2.1 Hardware Requirements

The computer hardware plays an important role, the server systems where we will install PLC software and Database software should have ssd hard disk and high speed processor.

Machine	**Ram and Hard disk**	**Processor**
2 Server System (Database and Machine)	32 gb ram, 512gb ssd hard disk	Intel Xeon® Silver 4310 @2.10 GHz
2 System for Operator (Infeed and Outfeed)	16gb ram, 256gb hard disk	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 1.38 GHz

3.2.2 Software Requirements

Operating System	Windows server 2019, Windows 10
Programming Language (Database)	MS SQL server

3.3 Results and Screenshots

SYSTORE is a “Windows based” application; it presents a multi-level menu, organized in different tabs basing on pages’ content and functionalities.

The operator interfaces with the management system via a dialog used to view, manage and control the system.

The general structure of the dialog is indicated below:

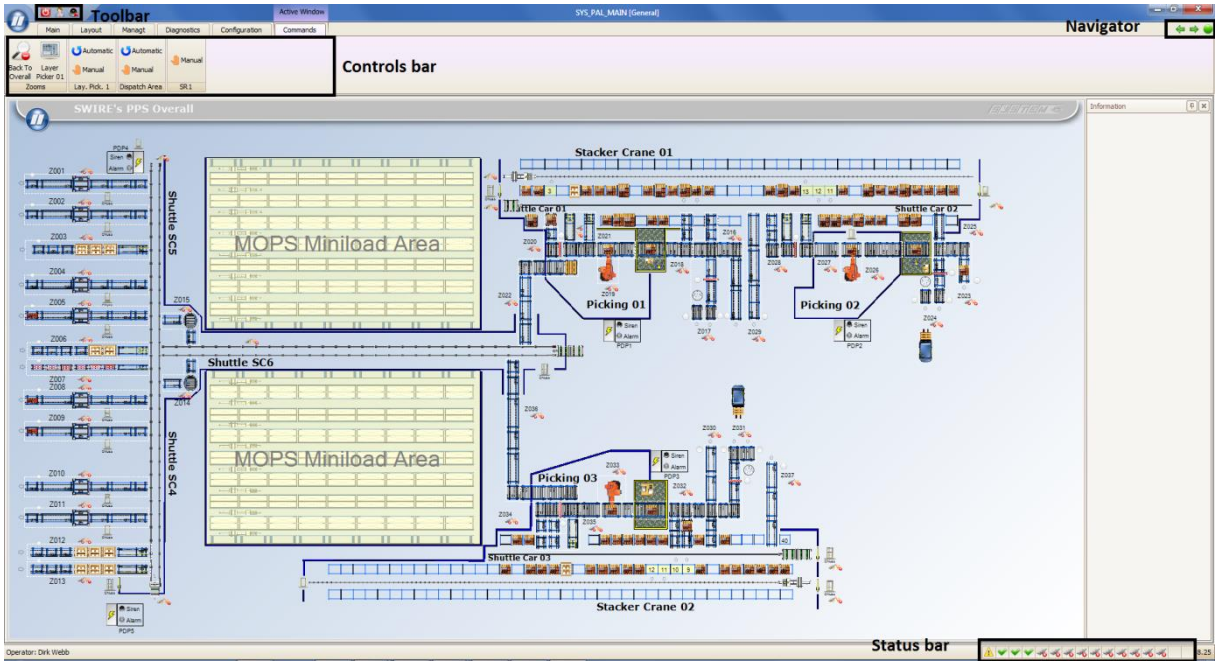


Fig 4 – Operator dialog

- **Tabs:** the tabs are list of buttons grouped by similar functionalities:
    - **Main:** The main tab in SYSTORE contains all the dialogs that need operation actions.
    - **Management:** The Management tab contains all the dialog to manage the main information of the WMS SYSTORE, like Articles, Loading unit, Orders, Missions…
    - **Diagnostic:** Diagnostic tab contains logs and history for the main activities of the plant
    - **Configuration:** Configuration tab contains dialogs to configure the parameters and components of the system.
    - **Commands:** commands tab contains the button of the active dialog
- The **Toolbar** offers all the management and navigation controls of the program.
- The **Control bar** offers all the control buttons related to the mimic page in use.
- The **Navigator** represents the exploration system of the program dialog.
- To zoom the item listed, click on the "+" symbol; click on the "-" symbol to go back to view the main folder normally.
- The **Status bar** shows certain operating conditions or events of the machine to be displayed despite the dialog in use.

The login page , we can define different users and assign them privileges based on the type of user like Operator, Supervisor or System user.

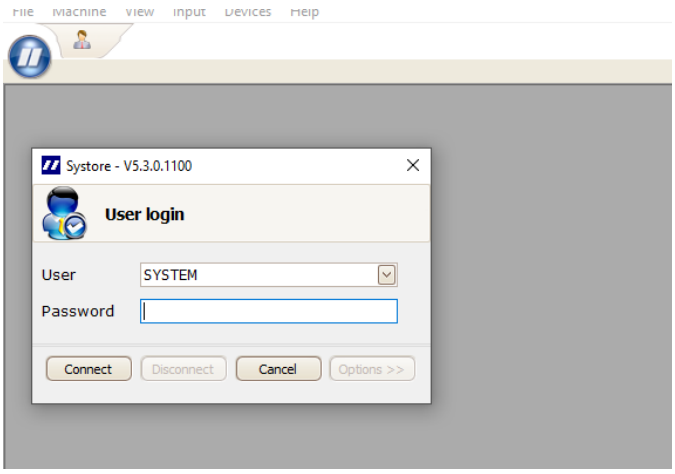


Fig 5: Login Page

The AS/RS consists of 3 primary parts in terms of Software development.

1. The Systore Application which will manage infeed/ outfeed and can track goods movement.

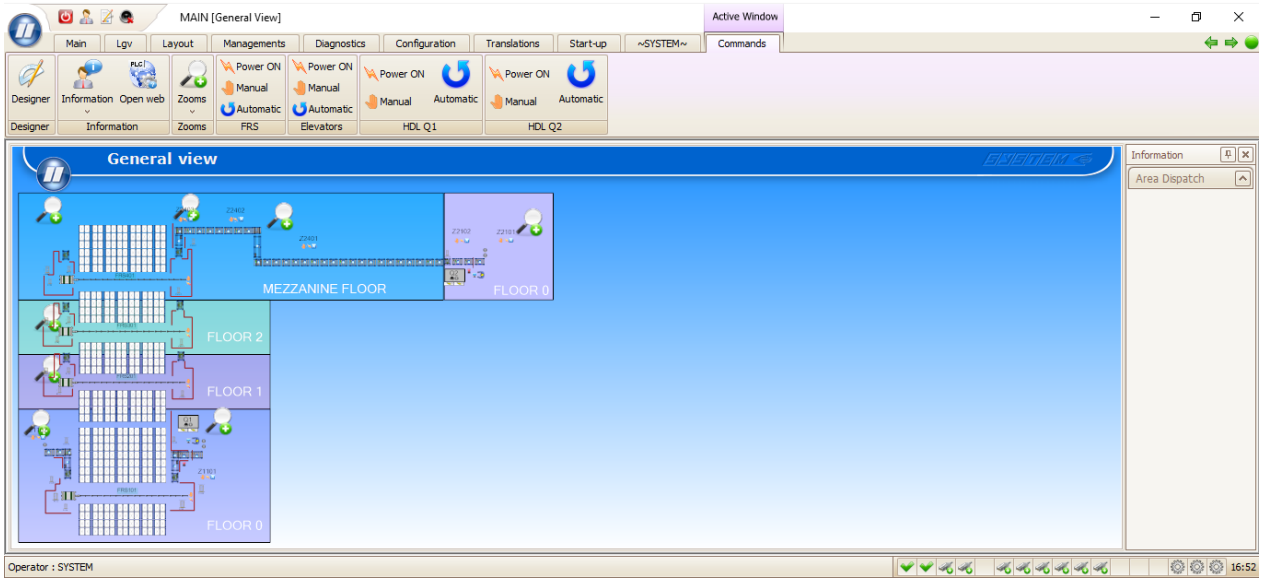
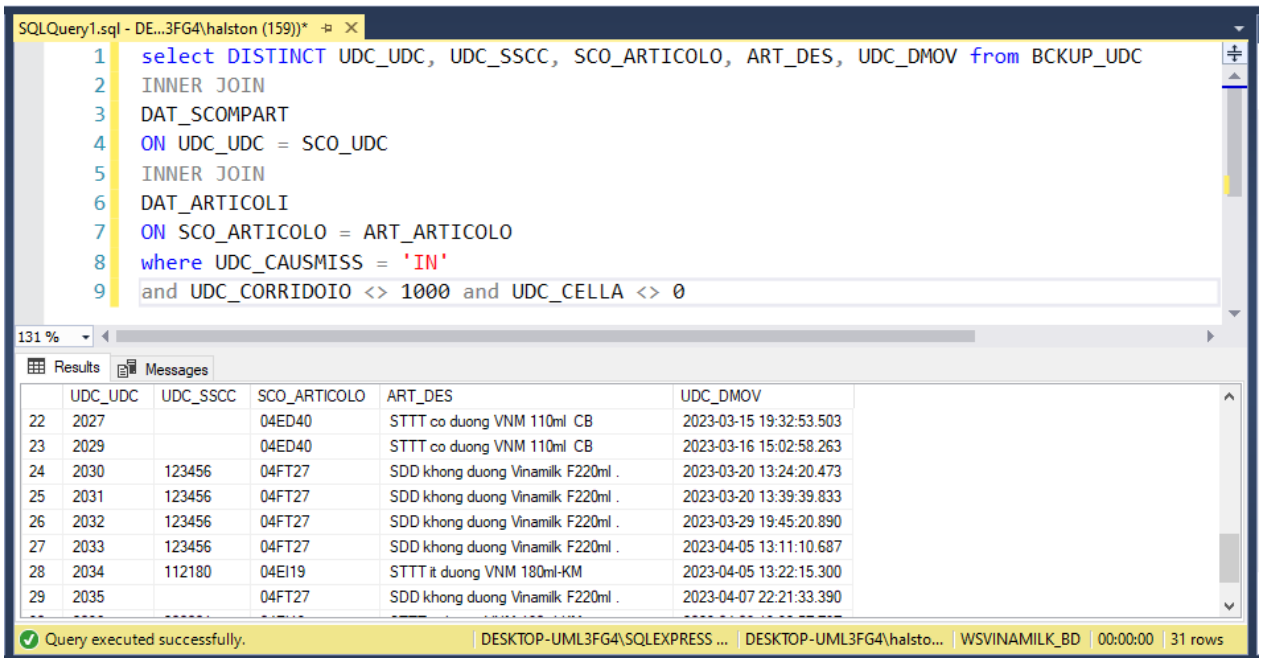


Fig 6: General View

2. The Database MS SQL server which will store the record of goods like Product name, Batch no, Weight, Expiry Date and Barcode.



	UDC_UDC	UDC_SSCC	SCO_ARTICOLO	ART_DES	UDC_DMOV
22	2027		04ED40	STTT co duong VNM 110ml CB	2023-03-15 19:32:53.503
23	2029		04ED40	STTT co duong VNM 110ml CB	2023-03-16 15:02:58.263
24	2030	123456	04FT27	SDD khong duong Vinamilk F220ml .	2023-03-20 13:24:20.473
25	2031	123456	04FT27	SDD khong duong Vinamilk F220ml .	2023-03-20 13:39:39.833
26	2032	123456	04FT27	SDD khong duong Vinamilk F220ml .	2023-03-29 19:45:20.890
27	2033	123456	04FT27	SDD khong duong Vinamilk F220ml .	2023-04-05 13:11:10.687
28	2034	112180	04EI19	STTT it duong VNM 180ml-KM	2023-04-05 13:22:15.300
29	2035		04FT27	SDD khong duong Vinamilk F220ml .	2023-04-07 22:21:33.390

Fig 7: SQL query to check goods details

3. The Simulator which will simulate dummy values which can be used for testing purpose before the actual site installation.

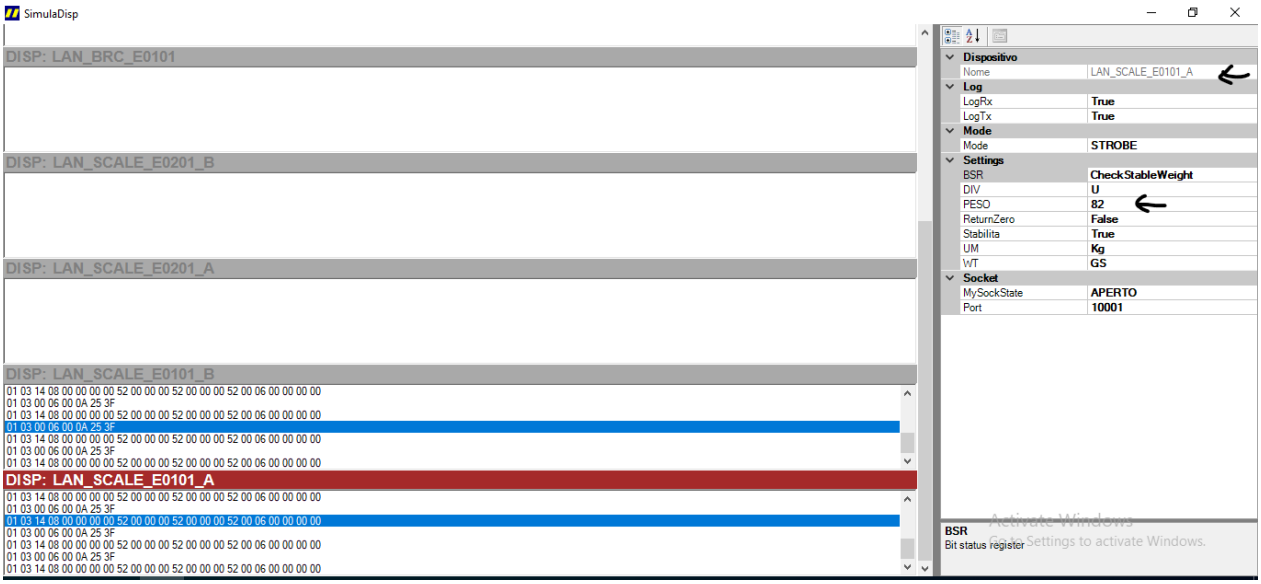


Fig 8: Simulator 1

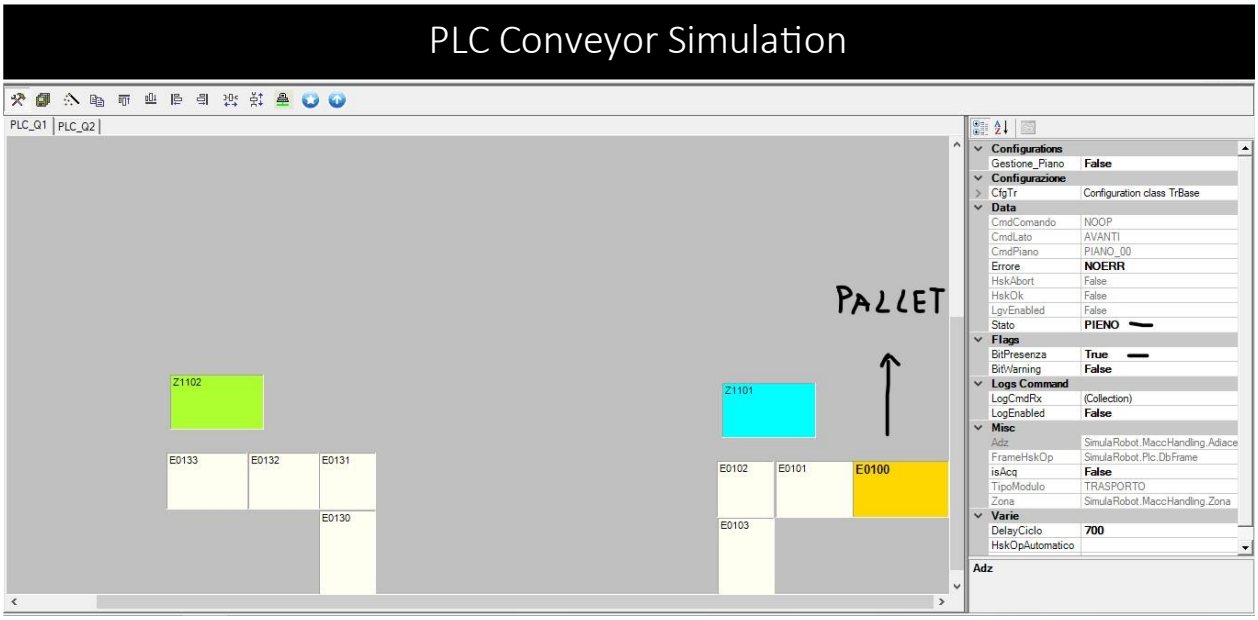


Fig 9: Simulator 2

A. Infeed

The Systore application gives us the live visuals of warehouse operations like product infeed, outfeed.

In the beginning of development, we use the simulator to create this test environment.

We can see a pallet present here, because of PLC Conveyor simulator (shown in previous page).

You can control the pallet movement then from manual option given.

We try to replicate the whole warehouse design which will be automated as shown in next slide.

In real scenario the goods are kept using the forklift on the conveyor.

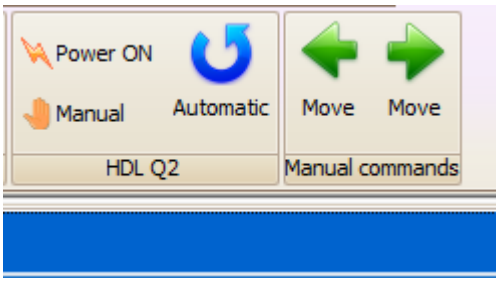


Fig 10: Manual Controls

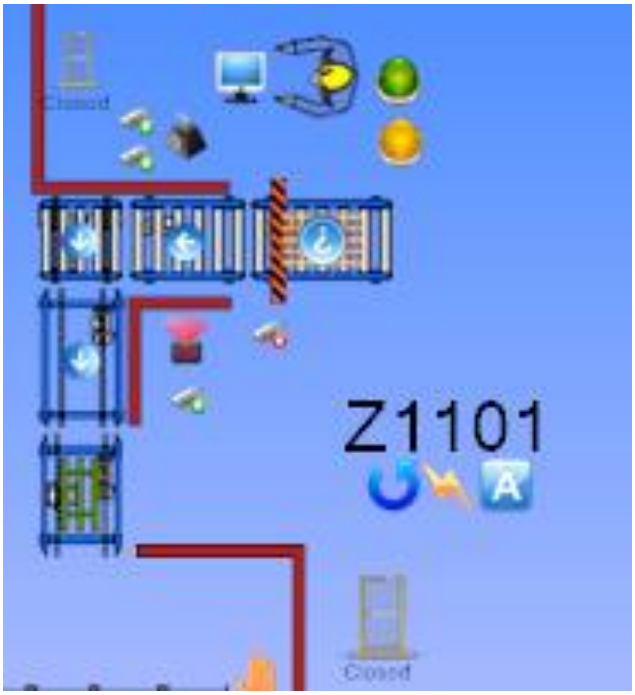


Fig 11: Infeed zone.

B. Outfeed

Now we have good stored inside when we get the order we need to outfeed (remove) it from inside the warehouse.

To do so we just go to Management -> Lu Location

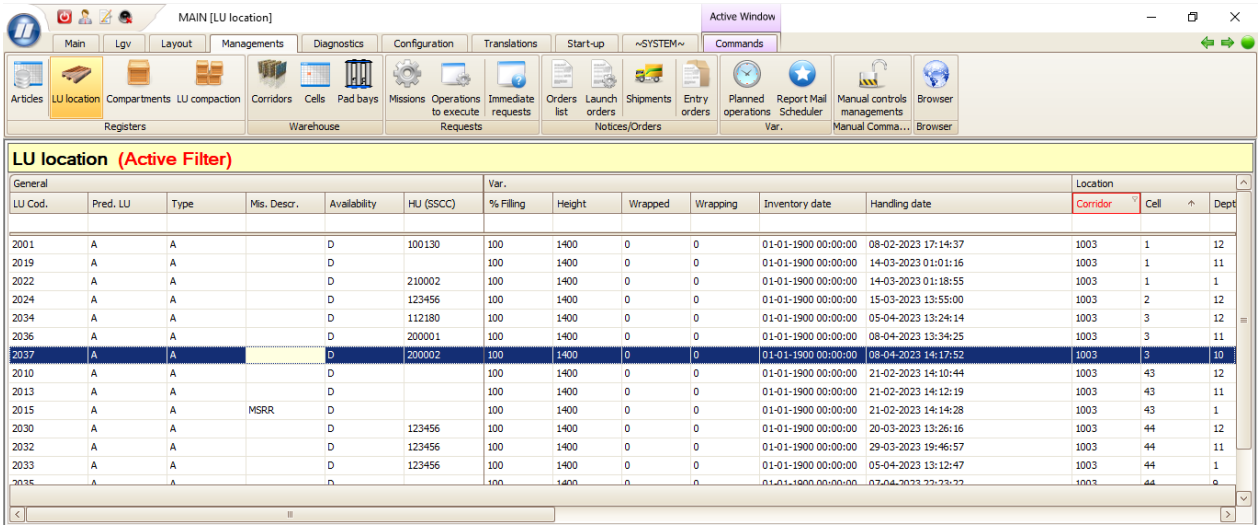


Fig 12: LU location

Select Immediate LU -> Select Destination Bay

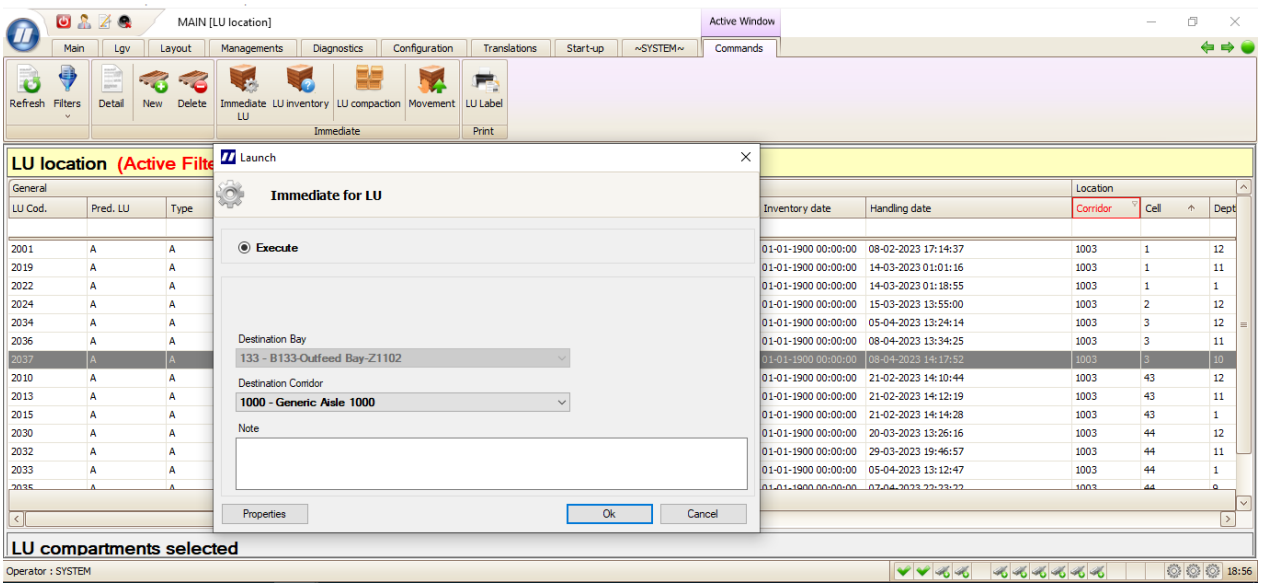


Fig 13: Immediate for LU

The pallet then moves towards its outfeed bay, in real scenario the forklift driver comes to this section and picks the pallet and move it towards generic area.

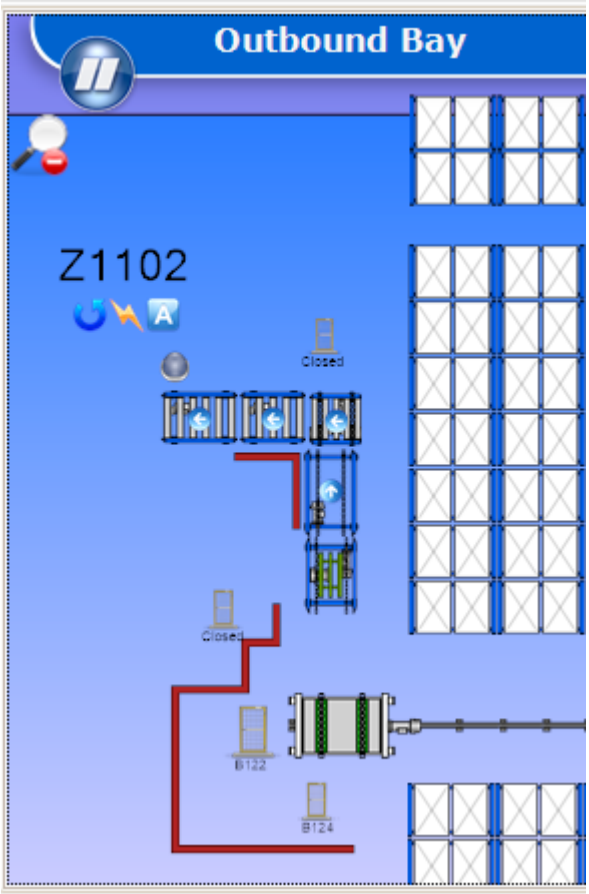


Fig 14: Outbound bay

C. Reject

If due to any reason the infeed fails, we can see the error details in the reject section like shown below.

Fig 15: Reject Bay with weight error.

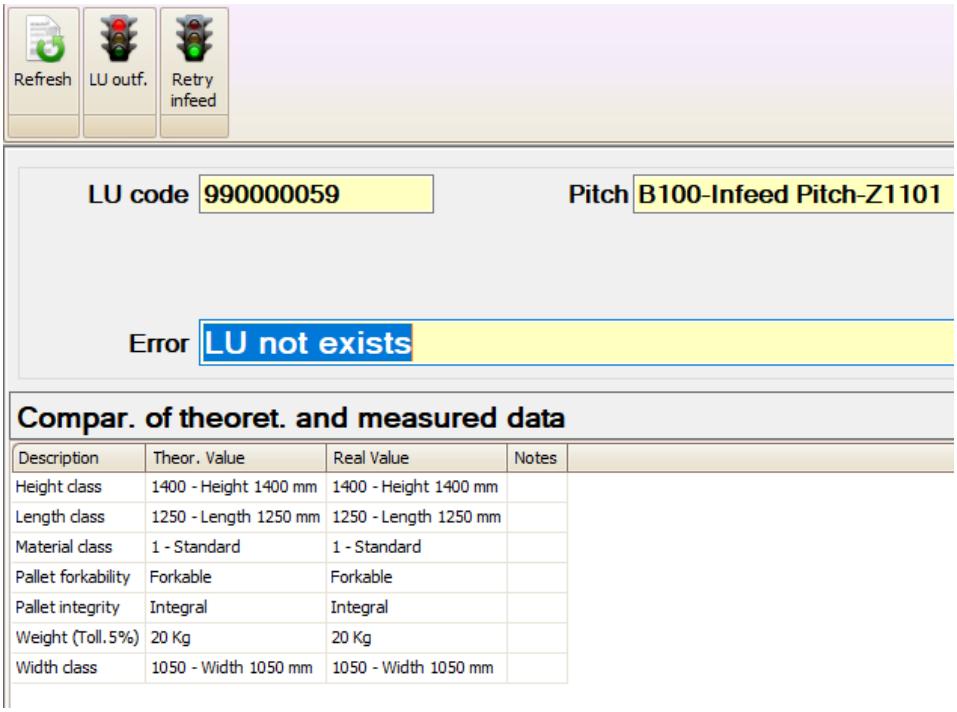


Fig 16: Reject Bay with LU not exists error

Corridors and Bays:

Configurazione Corridoi

Corridor	Corridor description	Enab. infeeds	Enab. outfeeds	Enab. booking	Area	Reorder active
1001	Storage 1001	☑	☑	☑	AS	☐
1002	Storage 1002	☑	☑	☑	AS	☐
1003	Storage 1003	☑	☑	☑	AS	☐
1004	Storage 1004	☐	☑	☑	AS	☐
20001	Shipment Corridor	☑	☑	☑	AG	☐
2001	Selective Rack(aisle 1)	☑	☑	☐	AM	☐
2002	Selective Rack(aisle 2)	☑	☑	☐	AM	☐
2003	Selective Rack(aisle 3)	☑	☑	☐	AM	☐

Fig 17: Corridors present on site

Bay configuration

Bay	Bay description	Enabling	Operation	Sched. Type	LU types	Max.outf. cap	Component	Bay uses	Active outfeed missions	Active operations to execute	Pitches disabled
100	B100-Infeed Bay-Z1101	☑	Infeed	Automatic		0	L2BASE_MAIN	,AVVIN,	1	1	
101	B101-Evaluation Bay-Z1101	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
103	B103-Decisional Bay-Z1101	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
133	B133-Outfeed Bay-Z1102	☑	Outfeed	Manual		3	L2BASE_MAIN	,ARTP,SC...	0	0	
200	B200-Infeed Bay-Z2101	☑	Infeed	Automatic		0	L2BASE_MAIN	,AVVIN,	0	0	
201	B201-Evaluation Bay-Z2101	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
202	B202-Decisional Bay-Z2101	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
203	B203-Decisional Bay-Z2101	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
296	B296-Decisional Bay-Z2403	☑	Infeed	Manual		0	L2BASE_MAIN		0	0	
3001	Picking bay	☑	Outfeed	Manual		288		,ORDP,P...	0	4	
4001	Shipping bay	☑	Outfeed	Manual		300		,SPEA,	0	0	

Fig 18: Bays on site

CHAPTER 4

Conclusion and Future Scope

In conclusion, automatic storage and retrieval systems (AS/RS) are a modern and efficient solution for managing inventory and maximizing warehouse space. These systems use advanced technology to automatically store, retrieve and move items within a warehouse, improving accuracy and speed while reducing labor costs.

AS/RS can be customized to fit the specific needs of a business, such as the type and size of items being stored and the layout of the warehouse. They offer a high degree of flexibility, allowing for easy adjustments to meet changing inventory needs.

Overall, AS/RS systems provide numerous benefits, including improved productivity, increased accuracy, and reduced labor costs. They are a smart investment for businesses looking to optimize their warehouse operations and stay competitive in today's market.

Future Scope, as per 2019 report only 4% warehouse/storage industries are automated, so bringing automation to more industries will showcase more problems and more R & D can be done , cost can be reduce and we try to make automation possible for mid-scale or smaller industries as well.

REFERENCES *(←IEEE ENDNOTE FORMAT) DOWNLOAD AT*

HTTPS://ENDNOTE.COM/STYLE_DOWNLOAD/IEEE/)

[1] final thesis.pdf (theseus.fi)

[2] (PDF) Automated Storage and Retrieval Systems: A Review on Travel Time Models and Control Policies (researchgate.net)

[3] System Logistics | Automated warehousing & picking, material handling, intralogistics

[4] Halston Automation (halstonsoftwares.com)

ANNEXURE

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

GR No.: **3480578** Roll no: **04**

1. Name of the Student: **BAROT PUNSHI RANSHI**
2. Title of the Project: **Automated Storage & Retrieval Systems**
3. Name of the Guide: **Prof. Rashmi Tiwari**
4. Teaching/Industry experience of the Guide: **years**
5. Is this your first submission? **Yes**

Signature of the Student: Signature of the Guide:

Date: ……. Date: …….

Signature of the Coordinator:

Date: ……